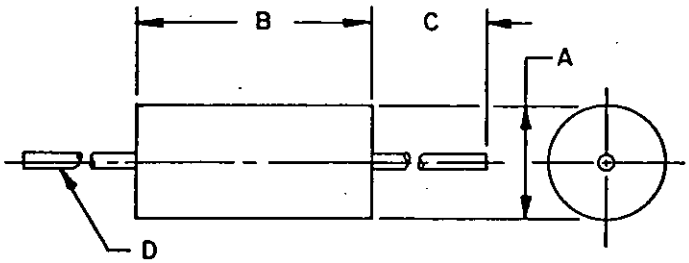
MILITARY SPECIFICATION SHEET

COILS, FIXED, RADIOFREQUENCY, MOLDED, MICRO-MINIATURE, MAGNETICALLY SHIELDED (IRON CORE-IRON SLEEVE), ESTABLISHED RELIABILITY

This specification is aproved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the coils described herein shall consist of this specification and the latest issue of MIL-C-39010.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.152 (3.86)	.172 (4.37)
B	.390 (9.91)	.430(10.92)
C	1.250(31.75)	1.626(41.30)
D	.023 (.58)	.027 (.69)

NOTE: Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.

REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Material: Iron core - iron sleeve.

Weight: 1.00 gram maximum.

Operating temperature range: -55°C to +105°C.

Dielectric withstanding voltage: Method 301 of MIL-STD-202 test voltage: 1,000 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202 (70,000 feet), test voltage 200 volts rms.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Inductance tolerance: See table I.

Q values: See table I.

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Self-resonant frequency: See table I.

Percent coupling: 3 percent maximum.

DC resistance: See table I.

Temperature rise: 15°C.

Terminal pull: 5 pounds.

Part number: M39010/02 (dash number from table I).

Revision letters are not used to denote changes due to the extensiveness of the changes.

TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz)	DC resistance (25°C) max (ohms)	Rated dc current (mA) 2/
A1R91**	.91	5	44	25	140	.07	1070
A1R0**	1.00	5,10	44	25	140	.07	1070
A1R1**	1.10	5	44	7.9	130	.10	895
A1R2**	1.20	5,10	44	7.9	130	.10	895
A1R3**	1.30	5	44	7.9	115	.12	815
A1R5**	1.50	5,10	44	7.9	115	.12	815
A1R6**	1.60	5	44	7.9	105	.14	755
A1R8**	1.80	5,10	44	7.9	105	.14	755
A2R0**	2.00	5	44	7.9	100	.19	650
A2R2**	2.20	5,10	44	7.9	100	.19	650
A2R4**	2.40	5	44	7.9	92	.28	535
A2R7**	2.70	5,10	44	7.9	92	.28	535
A3R0**	3.00	5	44	7.9	85	.35	490
A3R3**	3.30	5,10	44	7.9	85	.35	490
A3R5**	3.60	5	44	7.9	75	.40	450
A3R9**	3.90	5,10	44	7.9	75	.40	450
A4R3**	4.30	5	44	7.9	70	.55	380
A4R7**	4.70	5,10	44	7.9	70	.55	380
A5R1**	5.10	5	44	7.9	65	.72	335
A5R6**	5.60	5,10	44	7.9	65	.72	335
A6R2**	6.20	5	44	7.9	55	1.02	280

See footnotes at end of table.

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TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz)	DC resistance (25°C) max (ohms)	Rated dc current (mA) 2/
A6R8**	6.80	5,10	50	7.9	55	1.02	280
A7R5**	7.50	5	50	7.9	50	1.32	245
A8R2**	8.20	5,10	50	7.9	50	1.32	245
A9R1**	9.10	5	50	7.9	46	1.62	220
A100**	10.00	5,10	50	7.9	46	1.62	220
A110**	11.0	5	50	2.5	44	2.00	200
A120**	12.0	5,10	55	2.5	44	2.00	200

1/ The dash number will include two additional letters, (indicated by **). The first additional letter will indicated the inductance tolerance and the second additional letter will indicate the failure rate symbol (e.g., M, P, R, S) and will be added to the end of the dash number.

2/ The rated dc current is based on 90°C ambient temperature with a 15°C rise.

TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from initial measurement		Allowable % from specified minimum value in electrical characteristics (initial) table	
	Inductance	DC resistance	Self-resonant frequency	Q
	Percent			
Qualification inspection				
Group II	±5	---	---	-10
Group IV	±5	±(3%+.001 ohm)	1/	-10
Group V	±5	±(2%+.001 ohm)	1/	-10
Quality conformance inspection				
Group C				
Subgroup 1	±5	±(2%+.001 ohm)	1/	-10
Subgroup 3	±5	---	---	-10
Subgroup 4	±5	±(3%+.001 ohm)	1/	-10

1/ The self-resonant frequency shall be not less than the value specified in table I.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - AS, CG, MC, SH
Air Force - 19

Agent:
DLA-ES

Preparing activity:
Army - ER

(Project 5950-0603-2)

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